

Safety Data Sheet

Effective Date 01/04/2017

1. Identification

Material Name : Glycidyl methacrylate
2-Oxiranylmethyl 2-methylacrylate
2,3 epoxy-propyl methacrylate

Synonyms : GMA

CAS Number : 106-91-2

Product Use : Plastic and rubber modifier, Photosensitive resin,
Fiber treatment agent, Coating resin.

Manufacturer / Supplier : **Mitsubishi Chemical Corporation**

Address : 1-1, Marunouchi 1-chome, Chiyoda-ku, Tokyo, Japan

Section : MMA Planning Department

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(0900 – 1700 Japan office hours only).

2. Hazards Identification**GHS Classification:**

Health	Environmental	Physical
Acute Toxicity - Category 4 (oral)	Aquatic Toxicity -	Flammable Liquid -
Acute Toxicity - Category 3 (dermal)	(Acute) Category 2	Category 4
Eye Corrosion /Irritation- Category 1	Aquatic Toxicity –	
Skin Corrosion /Irritation - Category 1	(Chronic)	
Skin sensitization - Category 1	Not Classified	
Mutagenicity - Category 2		
Reproductive Toxicity - Category 2		
Target Organ Toxicity (Single) - Category 1 (respiratory system)		
Target Organ Toxicity (Repeated) - Category 1 (nervous system, blood vessel system, liver, kidney, respiratory system)		

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GHS Label:
Symbols:

Hazard Statements
DANGER!

Flammable Liquid.

Poisonous if inhaled.

Causes severe skin irritant and eye corrosion.

May cause allergic skin reaction.

May cause hereditary disease.

Suspected of damaging the unborn child or reproductive potential.

Suspected of damaging to respiratory system.

Suspected of damaging to nervous system, blood vessel system, liver, kidney, respiratory system by long time or repeated exposure.

Toxic to aquatic life.

Precautionary Statements

Obtain, read and understand the special instructions.

Wear designated respiratory protection, protective gloves, eye/face protection and protective wears.

Keep away from heat/sparks/open flame/high temperature objects.

No smoking.

Tightly close container.

Take precautionary measures against static discharge.

Use explosion proof electric apparatus / ventilation apparatus / lightning apparatus.

Use only non sparking tools.

Do not breathe mist/vapors.

Avoid environmental exposure.

Do not eat, drink or use tobacco when using this product.

Do not take out the soiled working wears from working area.

Wash hand sufficiently after handling.

Counter measures

Keep victim calm. Obtain medical treatment immediately.

Storage

Storage in tightly closed container at cool and well-ventilated area.

Storage at locked room.

Dispose

Contents and container are burned in designated incinerator or are entrusted to qualified waste trader.

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Another Hazards out of GHS classification

This material is very flammable. (flash point is 83°C)

High temperature or direct rays may cause polymerization. Polymerization followed exothermic heat and may cause explosive reaction.

3. Composition / Information on Ingredients

Component	CAS Number	Weight %
Glycidyl methacrylate	106-91-2	≥97%(normal state100%)
EINECS Number	TSCA	
203-441-9	Registration	

4. First Aid Measures

Eye: Eye irritation. Flush immediately with large amounts of water for at least 15 minutes.

Eyelids should be held away from the eyeball to ensure thorough rinsing.

If possible, remove contact lenses before flushing the eyes.

Get immediate medical attention.

Skin: Itching or burning of the skin. Immediately remove contaminated clothing and shoes. Flush the skin with plenty of running water at normal or lukewarm temperatures. Get immediate medical attention if the appearance of the skin change or if irritation persists.

Inhalation: Remove exposed person from source of exposure to fresh air. Wrap in a blanket or similar material to keep warm. Keep quiet. If breathing is weak or stops, loosen clothing, secure an open respiratory airway, and start cardiopulmonary resuscitation (CPR). Get immediate medical attention.

Ingestion: Get immediate medical attention. Do not induce vomiting unless directed by medical personnel. Doing so will increase the potential for harm. Should be allowed to wash the mouths with water, if the person himself wish. Do not place anything in the mouth if the person himself is unconscious.

Most important Sign and Symptoms: In case inhalation, may cause coughing, burning sensation, shortness of breathing, difficulty in breathing etc. Contact to skin may cause redness, burning of skin, pain, blisters. Contact to eye may cause redness, pain, losing eye-sight, deep burning. If swallowed may cause gastrospasm, abdominal pain, burning

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sensation, cataplexy.

Protection of rescuer: In case rescue the victims, wear suitable protective equipments.

5. Fire Fighting Measures

Suitable Extinguishing Media: Use dry chemical, foam, carbon dioxide or dry sand to extinguish fire.

Unsuitable Extinguishing Media: Do not use water in a jet. Doing so may be in danger the spread fire.

Special Hazards: Nothing special item.

Fire-fighting procedures: Keep persons not directly involved in fire-fighting out of the area of the fire. Use dry chemicals, a CO₂ extinguisher, dry sand etc. to extinguish a fire in its early stages. In case of large fires, deprive the fire of a source of oxygen using a foam extinguisher or by other means. Wear protective equipment during fire fighting.

6. Accidental Release Measures

Person protection measures (Protective equipments): Wear protective equipment and operate from upwind during clean up works. Isolate hazard area by rope and deny entry to unnecessary of unprotected personnel.

Environment protection measures: Do not discharge the spill into sewer pipes. If the spill involves small amounts, soak up the material with sawdust, waste cloth or sand, then collect the soaked materials into a container. For extensive spills, block the flow of liquid with earth or sand, cover the liquid surface with foam, then collect as much of the material into an empty container as possible. We recommend using spark proof tools.

Protects from secondary accidents: Promptly remove any flammable materials from the surrounding area. Prepare extinguishing media providing ignition.

7. Handling and Storage**Handling:**

- Do not use compressed air to charge, bleed or handle the product.
- If the product is to be handled indoors, provide good ventilation using a local exhaust system or other measures. (In case evaporating at 20 °C atmosphere may be contaminated and the concentration may arrive at poisonous level quickly.)
- Keep the product away from flames, sparks, strong oxidizers and high temperature

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substances.

- Handle the container gently; do not overturn, drop, strike or drag.
- Wear adequate protective equipment to prevent absorption through the skin, contact with the skin or eyes or inhalation of vapors. -- See Protective equipment.

Storage:

- Close the container tightly. Store in a cool, dark, well-ventilated place, away from direct sunlight.
- Do not store in the same area with organic peroxides
- If the product is to be placed in long-term storage (over three months), inspect the polymerization inhibitor concentrations for significant declines from initial concentrations as necessary.

If happen polymerization and heat up, the container should be transferred quickly to well ventilated wide area and cool down it by large quantity water. After that treat adequate method.

In case using bulk tank, store under 30°C (under 20°C is advisable). The space over liquid level in the tank should be controlled to low level oxidation concentration. This treatment may avoid the tank space going in flammable limits. The other side for avoiding polymerization, the oxidation concentration of tank space needs suitable level. So it will be desirable that the tank space is sealed by the gas that oxidation concentration is about 8%.

Recommended Material:

For containers use steel drum, UN Clean drum, Plastic coated drum.

8. Exposure Controls / Personal Protection

Control concentration: Not established.

Exposure Limits: Not established.

Engineering Controls: If the product is used indoors, seal the point of emission or install ventilating equipment or both. Install a safety shower, wash basin and eye bath near the handling area. Indicate the locations of these facilities clearly and prominently.

Personal Protective Equipment (PPE)

Respiratory Protection: Gas mask (to protect against organic gases), airline respirator (in atmospheres with high product vapor concentrations), air respirator, safety goggles or protective shield.

Hand Protection: Avoid skin contact. Wear oil-resistant gloves and boots.

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Eye Protection: Wear chemical safety goggles and face shield.**Skin and body Protection:** Additional protection may be necessary to prevent skin contact including use of apron, face shield, boots or full body protection.**9. Physical and Chemical Properties**

Appearance:	Clear, colorless liquid
Odor:	Ester weak odor
Threshold Limit Value of Odor	N.A.
Flashpoint:	83°C(181.4 °F) (open cup) 97°C (Creabuland open cup)
Boiling Point:	189°C(372.2 °F) @ 1013 hPa
Melting Point:	-65°C(-85 °F)
Auto-ignition Temperature	N.A.
Degradation Temperature	N.A.
Flammability Limit:	> 1.3vol% (Lower limit) <13%(upper limit)
Vapor Pressure:	1,333Pa 10mmHg (75°C)
Solubility in Water:	2.3g/100g Water
Distribution Coefficient (n-Octanol / Water) :	log Pow 0.81
Specific Gravity:	1.074 @25°C
Viscosity:	2.14cp/25°C
pH:	N.A.
Molecular Weight:	142.16
Refractive index (nD²⁰)	1.4482(25°C)

10. Stability and Reactivity**Stability:** A polymerization inhibitor has been added to the product to stabilize it under normal conditions of handling.**Hazardous Reactivity:** Polymerization followed exothermic heat and may cause explosive reaction.**Condition to Avoid:** Avoid to storage over the specified restrictions period or temperatures.**Incompatibility:** Incompatible with strong acids and peroxide. (Cause of polymerization)**Hazardous Decomposition Products:** N.A.

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11. Toxicological Information**Acute Oral Toxicity:** LD₅₀(Rat) 500 mg/kg---- (2)**Acute dermal Toxicity:** LD₅₀(Rabbit) 470 mg/kg---- (3)**Skin Corrosion / Irritation:** According to Skin Irritation test by rabbit at 4hrs , mild or middle level red spots, dropsy and mortification are observed within 14days.**Eye Corrosion / Irritation:** Eye Corrosion test by rabbit reported that moderate to sever irritation and corneal damage. Corneal damage did not heal within 7 days post-dosing. ----(4)**Respiratory Sensitization / Skin Sensitization:**

Respiratory Sensitization – No information.

Skin Sensitization – According to allergic reaction test by marmot is positive. ----(4)

– According to occupational exposure study, two positive reaction examples are reported.

Mutagenicity:

Germ Cell in vivo. Mutagenicity test (micronucleus test) – positive.----(3),(4),(5),(6)

Carcinogenicity: – No information..**Reproductive and Developmental Toxicity:**

According to generative studies, this material show decrement of child value and increase of respiratory embryo at doses which are maternally toxic. ----(2),(4),(5)

Target Organ Effects(Single): Inhalation test by animal, lungs, chest and breath are influenced. From those reports, target organ may be respiratory system.----(4)**Target Organ Effects(Repeated):** About animal some report show that nervous system, blood vessel system, liver, and kidney are damaged. Another report shows that nasal cavity organization and hypermorphosis of air duct are observed. From those reports, target organ may be central nervous system, blood vessel system, liver, kidney and respiratory system.----(4)**12. Ecological Information****Organism Toxicity/ degradability:** Pisces (cyprinodont) LC₅₀/96hr 2.8mg/L---(7)**Persistence / Degradability:** Readily biodegradable, Degradability by BOD is 94.0%.----(8)**Bioaccumulation:** Bioaccumulation is not expected. Log Kow = 0.81----(9)

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13. Disposal Considerations

Follow the instructions given in Handling and Storage (Sec.7). Observe general precautions applicable to flammable toxic liquids.

When incinerating the product, soak up the product using sawdust or similar materials, and burn in small increments. When disposing the products, should entrust to qualified waste trader. We recommend that the Waste Water including these products are treated by activation sludge equipments and are wasted to public space after confirming the quality. When wasting the containers used these products, wash these by water after confirming that the containers do not remain the products.

14. Transport Information

International Regulations : Follow UN classification

UN Classification : Class 8 (Corrosive substance) Second hazard Class 6.1(Poison·Toxic)

UN Number : UN 2922

Proper Shipping name : Other corrosive liquid (toxic)

Container Classification : III

Special Safety Measures : Avoid leak the products according to handle the container gently; do not overturn, drop, strike or drag. Watch the temperature of products, especially suddenly temperature raise.

Special Safety Measures and Conditions at Transportation : Follow the instructions given in Handling and Storage (Sec.7). Observe general precautions applicable to flammable toxic liquids.

First Aid Measurements Guide Number : 153P

15. Regulatory Information

Note: The latest regulatory information should be confirmed with the importer.

16. Other Information

Revision Indicator: Revised on 01/04/2017

Disclaimer: This SDS was prepared on the basis of materials, data and information available to us at the time of its writing. However, we make no warranty as to the accuracy

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of the data herein with respect to product content, physical and chemical properties thereof, or associated hazards. The precautions given herein apply to normal handling. For handling under special conditions, users are responsible for implementing adequate safety measures.

Reference

- (1) AVGIH (TLVs and BELs 2011)
- (2) The Ministry of the Environment of Japan (Environmental risk evaluation of chemicals) (Vol. 3 (2004)) (Vol. 4(2005))
- (3) Chemical Evaluation Research Institution (CERI) (Chemicals Safety and Hazards Data Book) (No.2000-38,2001)(No.2000-17,2002)
- (4) OECD SIDS Report (SIDS Initial Assessment Report)
- (5) The Ministry of Health, Labor and Welfare Test Report (Toxicity Test Report of chemicals) (Vol.5(1997))
- (6) NTP D.B.
- (7) The Ministry of the Environment of Japan (Ecological effect examination) (1996)
- (8) The Ministry of Economy, Trade and Industry of Japan (Chemical safety check data) (December 27,1991)
- (9) PHYSROP Database (SRC, 2005)